

***Monoclonal gammopathy of undetermined significance,
smoldering multiple myeloma, and curcumin: a randomized,
double-blind placebo-controlled cross-...***

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Monoclonal gammopathy of undetermined significance (MGUS) and smoldering multiple myeloma (SMM) represent useful models for studying multiple myeloma precursor disease, and for developing early intervention strategies. Administering a 4g dose of curcumin, we performed a randomised, double-blind placebo-controlled cross-over study, followed by an open-label extension study using an 8g dose to assess the effect of curcumin on FLC response and bone turnover in patients with MGUS and SMM. 36 patients (19 MGUS and 17 SMM) were randomised into two groups: one received 4g curcumin and the other 4g placebo, crossing over at 3 months. At completion of the 4g arm, all patients were given the option of entering an open-label, 8g dose extension study. Blood and urine samples were collected at specified intervals for specific marker analyses. Group values are expressed as mean $\pm$ 1 SD. Data from different time intervals within groups were compared using Student's paired t-test. 25 patients completed the 4g cross-over study and 18 the 8g extension study. Curcumin therapy decreased the free light-chain ratio (rFLC), reduced the difference between clonal and nonclonal light-chain (dFLC) and involved free light-chain (iFLC). uDPYD, a marker of bone resorption, decreased in the curcumin arm and increased on the placebo arm. Serum creatinine levels tended to diminish on curcumin therapy. These findings suggest that curcumin might have the potential to slow the disease process in patients with MGUS and SMM. Copyright © 2012 Wiley Periodicals, Inc.